

Tick life cycles

All ticks pass through egg, larva, nymph and adult. How many hosts they use, how long they feed and how many nymph stages they have depend on the family. This page shows three ticks common in South Africa, one for each type of life cycle, with the number of days spent in each stage.

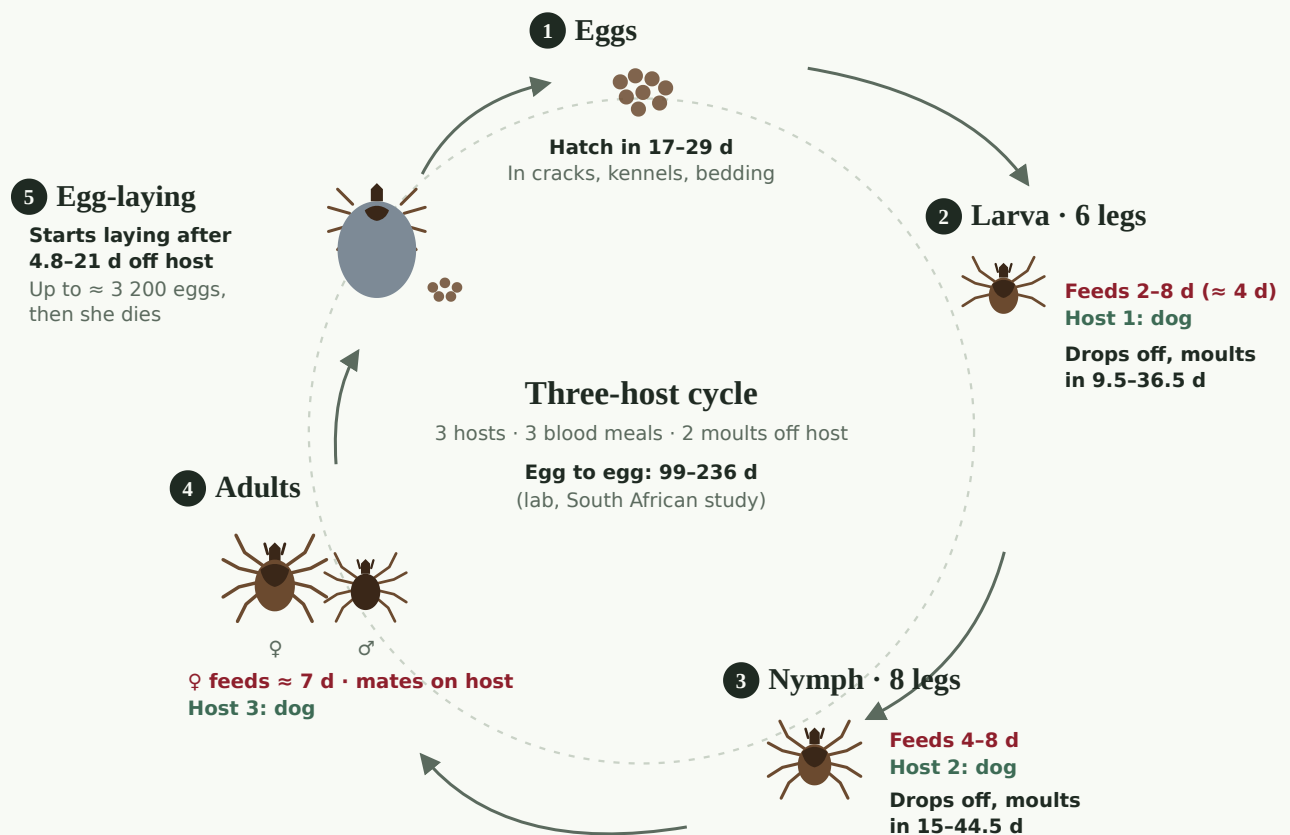
Red = time on the host, feeding **Black** = time off the host (moulting, egg-laying, hatching)

d = days · Lab ranges vary with temperature and humidity

HARD TICK · THREE-HOST

Brown dog tick (*Rhipicephalus sanguineus*)

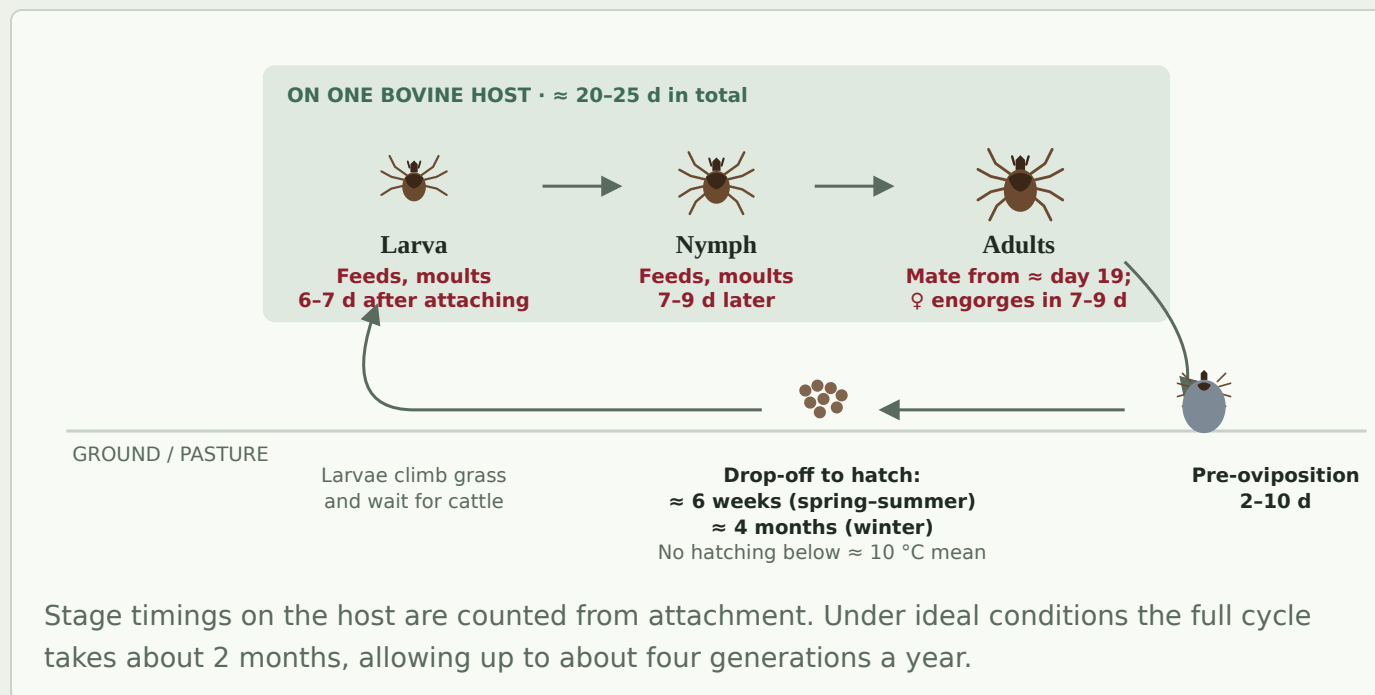
Found on dogs countrywide and often completes its whole cycle indoors, in kennels and houses. It drops off to moult between each stage and climbs back onto a new host, usually another dog.



Schematic, not to scale. The scutum (hard back shield) covers only the front of the female's back so she can swell, but the whole of the male's back.

African blue tick (*Rhipicephalus (Boophilus) decoloratus*)

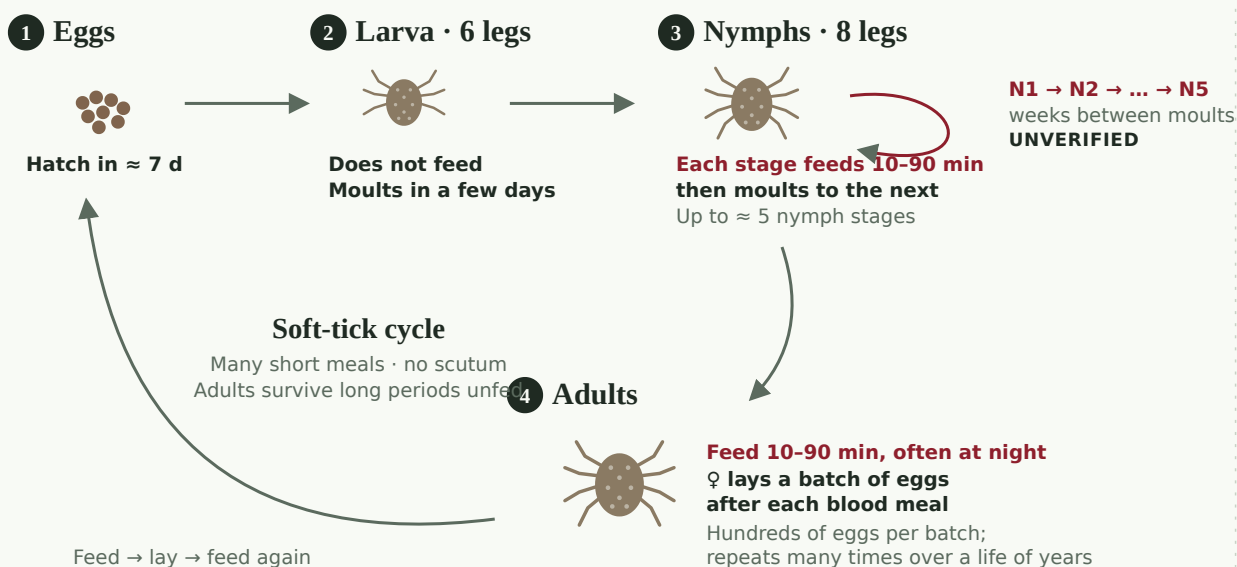
A major cattle tick in South Africa. Larva, nymph and adult all feed on the same animal and moult in place. Only the engorged female drops to the ground.



Eyeless tampan (*Ornithodoros moubata* group)

Lives in warthog burrows and animal shelters in northern South Africa, mainly in wildlife areas, and carries African swine fever virus. Soft ticks have no scutum and their mouthparts sit underneath the body. They hide in the host's shelter and feed quickly at night, for minutes instead of days.

ALL STAGES LIVE IN THE HOST'S BURROW OR SHELTER · they climb on only to feed



Number of nymph stages varies with blood-meal quality and source (reported 2-7 moults).
Feeding times are from lab membrane feeding.

Days in each stage

STAGE	BROWN DOG TICK (THREE-HOST)	AFRICAN BLUE TICK (ONE-HOST)	EYELESS TAMPAN (SOFT)
Egg → hatch	17-29 d mean 21.2 d, lab	≈ 6 weeks from drop-off (summer) ≈ 4 months in winter	≈ 7 d
Larva feeding	2-8 d typically ≈ 4 d	Moults 6-7 d after attaching on the host	Does not feed
Larva → nymph moult	9.5-36.5 d off host	Included above	A few days
Nymph feeding	4-8 d	Moults 7-9 d later on the host	10-90 min per stage
Nymph → adult moult	15-44.5 d off host	Included above	Weeks per stage UNVERIFIED
Adult female feeding	≈ 7 d	7-9 d, then drops off	10-90 min per meal
Pre-oviposition	4.8-21 d	2-10 d	Not found UNVERIFIED
Whole cycle	99-236 d lab, South Africa	≈ 2 months (ideal) up to ≈ 4 generations a year	Months to years adults live for years

Ranges come from different studies and conditions, so the stage totals do not add up exactly to the whole-cycle figures. Cold, dry weather lengthens every off-host stage.

Hard ticks vs soft ticks

	HARD TICKS (IXODIDAE)	SOFT TICKS (ARGASIDAE)
Back shield	Hard scutum	None; leathery, wrinkled body
Mouthparts	Visible from above, at the front	Underneath the body, hidden from above
Nymph stages	One	Several
Feeding time	Days per stage	Minutes to hours per meal
Meals as adult	One; female lays once, then dies	Many; female lays after each meal
Where they live	On the host while feeding; vegetation or kennels between meals	In the host's burrow, nest or shelter

Sources

- Jacobs, Fourie & Horak (2004). A laboratory comparison of the life cycles of the dog ticks *Haemaphysalis leachi* and *Rhipicephalus sanguineus*. *Onderstepoort J Vet Res*. [PubMed 15185571](#)
- Laboratory biomorphology of *Rhipicephalus sanguineus*, *Journal of Bioresource Management* 10(1). [Wright State CORE Scholar](#)
- Fecundity, oviposition and egg incubation period of female *R. sanguineus* in Indonesia. [JSciMed Central](#)
- ESCCAP UK: [Life cycle of the brown dog tick](#)
- Farmspace: [The African blue tick](#)
- Phalatsi et al. Larval biology of *R. (B.) decoloratus* in Free State Province. [Onderstepoort J Vet Res](#)
- Londt: [Prehatch period and larval survival of *B. decoloratus* under natural conditions in the Transvaal](#)
- [Ornithodoros moubata](#) (African hut tampan; the eyeless tampan), *Trends in Parasitology* (2025)
- [Maintaining the life cycle of *O. moubata* using in vitro feeding](#)
- [ScienceDirect Topics: Ornithodoros moubata](#)